

scope for people trained in optical communication. He says "The optical communication industry is well positioned for growth in the days to come, despite near-term uncertainty in the market—a phenomenon that keeps industry *panchits* on their toes. However, employees have nothing to fear, for growing competence levels ensure that there are plenty of opportunities for aspirants. There are very good opportunities in optical fibre cable ducting, fibre blowing, splicing and other related activities. Also, there are increased opportunities in optical distribution network surveying and designing."

Qualification

"Unfortunately, no Indian university/institute produces undergraduate-level optics and optoelectronics engineers! Therefore one needs to earn a post-graduate-level degree with specialisation in this subject," informs Das.

Currently, Department of Electrical Engineering at IIT Madras has an M.Tech programme in photonics. IIT Delhi offers M.Tech in applied optics within the curriculum of Physics Department. IIT Kharagpur too has an M.Tech programme in the area of fibre and integrated optics.

"Besides, there are M.Tech programmes in Kolkata University and Kochin University of Science and Technology. Sensing the future prospects, many private engineering colleges have recently started degree programmes in optoelectronics," adds Das.

A BE or B.Tech. in electronics and communication engineering with optical communication as a core subject, M.Sc physics with optoelectronics or M.Sc in nanotechnology is important to kickstart a career in this industry.

Mittal says, "For field engineering in optical communication, the aspirant must be a degree/diploma holder in electronics. Any exposure to optical communication will be an added advantage."

Talking about designing, he says, "For optical distribution network design, you must have a degree along

Companies to look out for

- OSI Optoelectronics, Hyderabad
- Optoelectronics Factory, Dehradun
- Opto Electronics Pvt Ltd, Bengaluru
- SFO Technologies, Cochin
- Satellite Optical Technologies Ltd, Pune
- Tejas Networks India Ltd, Bengaluru
- Opto Circuits (India) Ltd, Bengaluru
- Kwalify Photonics Pvt Ltd, Hyderabad

with experience in optical communication. In-depth knowledge of product development life-cycle management from concept through to production is not an absolute necessity. However, sound knowledge and background in design and development process are desirable."

Pay package

"Recently, I came across an article stating that the global optical communication industry maintained stable growth in 2011, and the total demand rose by almost as much as 15 per cent year-on-year in 2011," says Mittal.

He continues, "The demand remained strong in 2012, with the growth rate of nearly 6 per cent. The trend indicates a promising future for the industry—a positive sign for those aspiring to make a career in the same. Freshers would be taking up roles in the field for optical ducting, splicing, etc in the beginning years, gradually taking up complex and better paying roles according to their skill and level of understanding in the field."

The pay package varies according to the role of an employee, roughly ranging from Rs 250,000 to Rs 400,000 for a fresher in optical communication.

Skills required

Optical communication industry thrives on skills and experience of the employees.

Mittal says, "While most insiders claim that most of the skills are acquired in the field, a bare minimum of skills and knowledge of the job is a must for aspirants. Quite often, the requisite amount of skill comes during initial training while pursuing a degree/diploma course. Basic knowledge

of optical communication is desirable but not always compulsory."

Talking about skills, Grover says, "Knowledge of communication engineering and semiconductor technology fundamentals, protocols/standards for optical networks, and higher-level languages (C++ and Java) is important."

Short-term courses

Mittal says, "There are no visible parameters of skills that one requires to make a career in the optical communication industry. However, there are short-term courses in optical ducting and splicing, which are an added advantage, of course, but again not compulsory. Otherwise, one can learn these during the initial training."

Grover adds, "A short-term course in planning, designing and testing of optical communication links and networks would be great."

In the future

"The optical industry is huge, largely dominated by display devices like LEDs and LCDs. But as the manufacturing base for these devices is very limited in India, there is limited scope for engineers in optical display manufacturing," says Mittal.

He adds, "A near-stagnant market has converted into one of the fastest growing industries, thanks to greater attention to quality and strong advertising campaigns."

Nevertheless, in global terms, the industry in India remains underdeveloped even today. "The high duty tariff on products is acting as a serious deterrent for major business players, encouraging unorganised sector (grey market) growth," says Mittal.

To promote optoelectronics sector as a career option, "Conducting awareness programmes on the importance of optoelectronics sector at university/industry level and collaboration of optical society of India and industry/university/R&D centres are some of the desirable initiatives," says Grover. ●

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